Break-even

Exam code: 9BS0

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Using the contribution to calculate the break-even point

- The contribution value is used to calculate the break-even point
- The break-even point is where **total revenue earned for a product is exactly equal to its total costs** and where the business is making **neither a profit nor a loss**
- It is calculated using the formula

$$\text{Break - even point} = \frac{\text{Fixed costs}}{\text{Contribution}}$$

- The break-even point is expressed in **units** (e.g. the number of scented candles)
- Identifying the break-even point allows a business to understand **how many items it needs to produce and sell** to cover all costs before it starts to make a profit
  - Each subsequent unit sold past this point will generate profit for the business

Worked Example

Cost and revenue data for Canterbury Glamping

Cost/revenue	Cost (£)
Revenue per pod per night	95
Variable costs per pod per night	19
Annual fixed costs	55,000

Using the information in the table, calculate how many pods need to be occupied each month for Canterbury Glamping to break even. [4]

Step 1: State the formula to calculate the break-even point

$$\text{Break - even point} = \frac{\text{Fixed costs}}{\text{Contribution}} \quad [1]$$

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$$= £95 - £19$$ [1]

$$= £76$$

Step 3: Apply the formula to calculate the break-even point

$$= \frac{£55{,}000}{£76}$$ [1]

$$= 723.68$$

Step 4: Always round up to the nearest whole number because only whole products can be sold

$$= 724 \text{ camping pods}$$ [1]

Margin of safety

- The margin of safety is the **difference** between the actual level of output of a business and its **break-even level of output**
- The margin of safety can be calculated using the following formula:

$$\text{Margin of safety} = \text{Actual level of output} - \text{Break-even level of output}$$

Worked Example

The cost, sales and revenue for an electric bicycle manufacturer are presented in the table below.

Annual fixed costs	£42,000
Selling price per unit	£750
Variable cost per unit	£350
Number of units sold	240

Using the data, calculate the margin of safety. You are advised to show your workings.

Step 1: Calculate the contribution

$$= £750 - £350$$

$$= £400$$

Step 2: Calculate the break-even point

$$= \frac{£42{,}000}{£400}$$

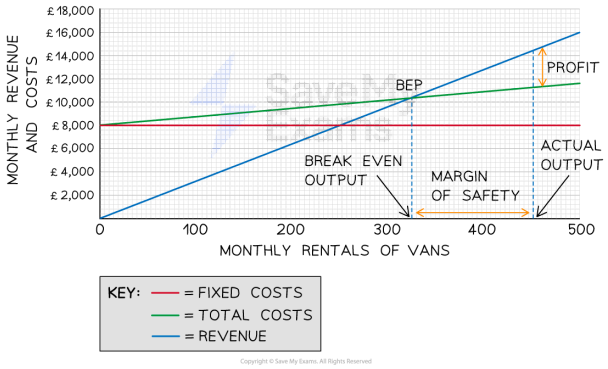
$$= 105 \text{ units}$$

Step 3: Calculate the margin of safety

Interpretation of break-even charts

- A break-even chart is a **visual representation** of the break-even point and is used to identify the following
  - **Fixed costs, total costs and revenue** over a range of output
  - The **break-even point**, where total costs are equal to revenue
  - The **profit or loss** made at each level of output
  - The **margin of safety**

An example break-even chart



The break-even chart for A2B Limited shows that at 324 units, the total revenue = the total costs

Diagram analysis

- Fixed costs do not change as output increases
  - A2B's fixed costs are £8,000, and these do not change whether the business produces zero units or 500 units

- Total costs are made up of fixed and variable costs
  - At zero units of output, they are made up exclusively of fixed costs
  - At 500 units, the total variable costs equate to £11,800
  - This line **slopes upwards** because total variable costs increase as output increases

- The revenue line also slopes upwards
  - At zero units of output, the revenue is £0
  - At 500 units, the total revenue equates to £16,000
  - Revenue will increase with the output
  - The line will **slope more steeply than the total costs** and will **cross the total costs line** at some point

- The point at which the total costs and the revenue lines cross is the **break-even point**
  - The break-even level of output for A2B is 324 units

- The profit made at a specific level of output can be identified as the **space between the revenue and total cost lines**
  - In this instance, the profit made at 450 units of output is £14,400 – £11,250 = £3,150

Limitations of break-even analysis

The limitations of break-even analysis include the assumption that all output is sold

Examiner Tips and Tricks

When assessing break-even analysis, ensure that you explain why it has an important internal planning role, but don't forget that it has a significant external role as well.

Break-even analysis should be included in a business plan when a business is trying to secure external finance. Businesses looking to borrow money or attract investors seeking to manage their risk should take care to model the break-even point, margin of safety and level of profit (or loss) at different levels of output and prepare themselves to be scrutinised on the figures.

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